

Can Large Language Model Priors Speak Multi-Omics? Textual Knowledge Constraints for Transcriptomic Clustering and Cancer Subtyping

Jiale Liu,¹ Jinli Zhang,^{1,*} Zheng Chen,² Zongli Jiang,¹ Ziwei Yang³ and Junzhe Jiang¹

¹College of Computer Science, Beijing University of Technology, 100 Pingleyuan, Chaoyang District, Beijing 100124, China

²The University of Osaka, 1-1 Yamadaoka, Suita, Osaka 565-0871, Japan

³Kyoto University, Yoshida-honmachi, Sakyo-ku, Kyoto 606-8501, Japan

*Corresponding author. lz73798@gmail.com

Abstract

Motivation: Transcriptomic clustering supports patient subtyping and cell-state discovery, but datasets combine sparse, batch-affected high-dimensional expression with limited and imbalanced labels, while generic pretrained embeddings are not optimized for clustering.

Results: We present STCKLA, a semi-supervised framework that combines foundation-model representations, label-guided learning and Qwen-assisted biological priors. STCKLA extracts class-associated candidate genes from the analysed dataset, then uses curated pathway, regulatory, tissue and marker evidence to construct signed marker modules with Qwen assistance. These modules help retain informative genes and constrain the learned embedding. Across two bulk RNA-seq and five single-cell RNA-seq benchmarks, STCKLA achieved the best overall ACC, NMI, ARI and PUR among evaluated semi-supervised, unsupervised, LLM-derived and foundation-model baselines. Component ablations and marker-module analyses supported the benefit of combining clustering objectives with biological priors.

Availability and implementation: Source code and workflows are available at <https://github.com/AIDataLab/STCKLA-Textual-Knowledge-Constraints-for-Transcriptomic-Clustering-and-Cancer-Subtyping>. The archived software version is available at Zenodo (DOI: 10.5281/zenodo.22976359).

Contact: lz73798@gmail.com.

Supplementary information: Supplementary data are available at *Bioinformatics* online.

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1. Introduction

Transcriptomic clustering is a central step for identifying cell populations and disease-associated molecular subtypes from expression profiles. It precedes marker-based interpretation in scRNA-seq and supports subtype discovery in patient-level transcriptomic cohorts. However, clustering is sensitive to the feature space in which samples are compared. Sparse scRNA-seq measurements and heterogeneous bulk profiles require representations that suppress nuisance variation while preserving coordinated expression signals defining the target biological partition.

Recent clustering models seek to improve this representation before or during optimization. Graph and attention-based approaches use cell-cell or gene-cell relations to refine latent structure (Cheng and Ma, 2022). Contrastive approaches define agreement between cells or augmented views to refine latent structure (Wan et al., 2022; Qi et al., 2026). Topology-aware extensions further preserve neighbourhood organization during clustering (Zhou et al., 2026; Meng et al., 2025). Semi-supervised approaches incorporate partial labels or pairwise constraints to reduce ambiguity in the target partition (Wang et al., 2023; Qiu et al., 2024). These methods demonstrate the value of task-aware objectives, but their performance still depends on pseudo-labels, reconstructed counts, neighbourhood graphs or augmentations estimated from the same target data. Sparse labels can indicate some desired distinctions, but they do not specify the molecular programmes that make those distinctions biologically coherent.

Transcriptomic foundation models provide a complementary starting point. scGPT and scFoundation demonstrate that

large-scale pretraining can encode transferable single-cell representations (Cui et al., 2024; Hao et al., 2024). CellFM and RegFormer further extend this direction through large-scale cellular pretraining and regulatory-structure-aware modelling (Zeng et al., 2025; Hu et al., 2026). Nevertheless, the objectives used for pretraining, such as rank prediction, masked-expression recovery or generative modelling, are not identical to the objective of forming compact clusters in a new target study. Benchmarking studies have reported gaps between annotation-oriented performance and clustering quality for several foundation-model embeddings (Xu et al., 2026; Kedzierska et al., 2025). This suggests that foundation representations often require task-specific adaptation, especially when the target tissue, disease state or sampling unit is under-represented in pretraining data.

Biological prior knowledge can help specify which expression patterns should matter for a particular clustering task. Pathway and regulatory resources such as Reactome and TRRUST connect gene-level signals to curated biological processes and transcriptional regulation (Gillespie et al., 2022; Han et al., 2018). GTEx and PanglaoDB provide complementary tissue and cell-identity context (GTEx Consortium, 2020; Franzén et al., 2019). Context-aware regulatory networks and sparse knowledge-guided feature weighting further show that biologically contextual information can improve downstream prediction when it is incorporated during learning (Tsutsui et al., 2026; Kim and Altman, 2025). Large language models offer an additional interface for organizing heterogeneous biological evidence. GenePT and scELMo show that language-model-derived representations can encode gene- or cell-level semantics

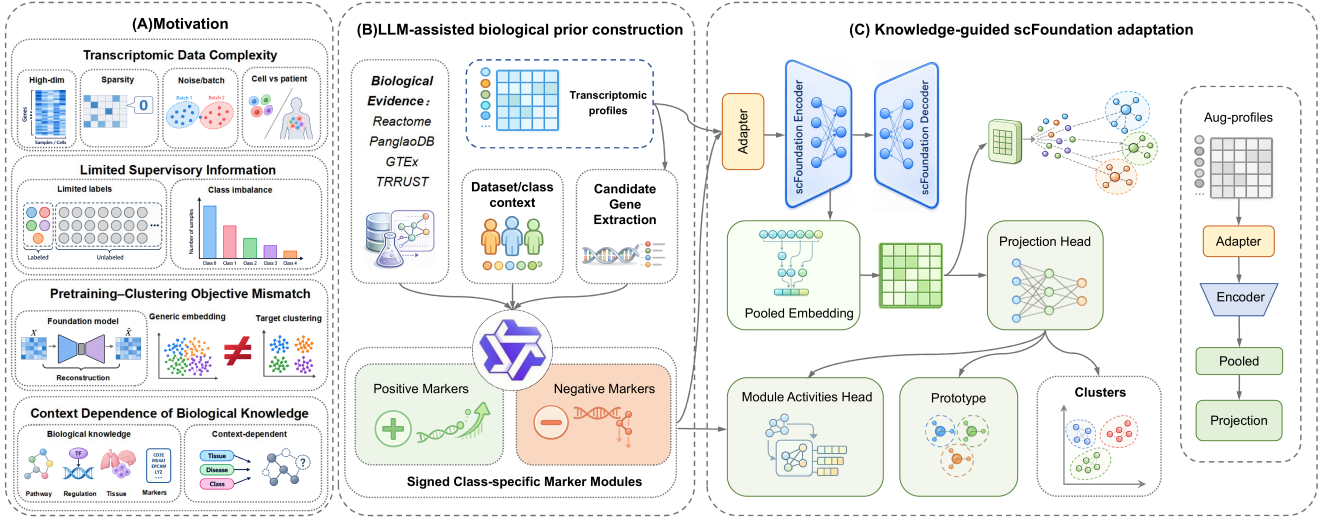


Figure 1 Overview of STCKLA. (A) Motivation for prior-guided transcriptomic clustering. (B) Qwen-assisted construction of signed class-specific marker modules from target-data candidates, biological evidence and class context. (C) Knowledge-guided scFoundation adaptation using residual adaptation, module-guided visible-budget token selection, activity alignment and clustering-oriented objectives. The dual-path integration of the Qwen-derived priors is detailed in Figure 2.

(Chen and Zou, 2025; Liu et al., 2026). Literature-grounded gene prioritization and enrichment summarization further illustrate how language models can connect genes to functions, diseases and pathways (Khan et al., 2025; Brinkmann et al., 2026). However, the integration of language-model-assisted biological reasoning with the latent geometry of semi-supervised transcriptomic clustering remains underexplored. A key issue is how LLM-organized biological evidence becomes consequential for target-cohort representation learning. Figure 1A summarizes these challenges, including complex transcriptomic data, limited supervision, pretraining-clustering objective mismatch and context-dependent biological knowledge.

These observations define the methodological gap addressed here: limited labels provide task information, foundation models provide transferable transcriptomic structure and biological knowledge provides contextual interpretation, but these sources are rarely coupled during representation adaptation for clustering. We introduce STCKLA, a semi-supervised transcriptomic clustering method guided by prior biological knowledge with large language model assistance. STCKLA makes the following contributions:

- It uses Qwen to organize target-data-supported candidate genes and curated evidence into class-specific biological hypotheses.
- It translates these hypotheses into module-level signals that are evaluated within representation learning.
- It adapts scFoundation with reconstruction, supervised classification, prototype learning, cross-view consistency and knowledge-alignment objectives so that labelled and unlabelled profiles are organized in a common clustering space.
- It evaluates the resulting model on patient-level and single-cell transcriptomic benchmarks, with ablation, module-validation and supplementary robustness analyses designed to test the role of knowledge-guided adaptation.

2. Methods

2.1. Prior-guided STCKLA architecture

Let $\mathbf{X} \in \mathbb{R}^{N \times G}$ denote a transcriptomic matrix with N profiles and G genes. Exposed labels index $\mathcal{L}$, the remaining profiles form $\mathcal{U}$, and C is the number of classes in the dataset. STCKLA learns a common clustering representation shaped by labels, pretrained structure and context-specific biological priors.

STCKLA organizes class-associated candidate genes into signed marker modules with Qwen and curated evidence, calibrates their weights, activities, reliability and gene priorities from the target cohort, and adapts scFoundation with reconstruction, supervised classification, prototype learning, consistency regularization and knowledge alignment. This progression treats Qwen output as class-specific biological hypotheses whose relevance is evaluated within target-cohort representation learning. Figure 1B shows module construction and Figure 1C their use during adaptation.

2.2. Contextual marker-module priors

Data-anchored marker nomination. For each class c and gene g , STCKLA first computes a class-versus-rest standardized effect,

$$d_{cg} = \frac{\bar{x}_{cg}^{\text{in}} - \bar{x}_{cg}^{\text{out}}}{s_{cg}^{\text{pool}}}. \quad (1)$$

where $\bar{x}_{cg}^{\text{in}}$ and $\bar{x}_{cg}^{\text{out}}$ are transformed mean expression values of gene g inside and outside class c , and s_{cg}^{pool} is the corresponding pooled variability. Large positive and negative values define up- and down-regulated candidate sets. Candidate nomination anchors Qwen-assisted module construction in cohort-specific expression signals.

Biological-context module assembly. Candidates are linked to Reactome pathway (Gillespie et al., 2022), TRRUST regulatory (Han et al., 2018), GTEx tissue (GTEx Consortium, 2020) and PanglaoDB marker evidence (Franzén et al., 2019). Qwen3.5-9B (Qwen Team, 2026) receives the candidates, evidence summaries and class descriptions, and returns selected positive

and negative markers, excluded candidates and a concise rationale. For class c , $\mathcal{M}_c$ denotes its signed marker module:

$$\mathcal{M}_c = \mathcal{M}_c^+ \cup \mathcal{M}_c^- \quad (2)$$

Qwen organizes module membership and direction from the evidence and class context, while target-cohort statistics and exposed labels quantify the resulting modules for downstream learning. This design couples context-aware biological organization with cohort-specific calibration. Complete programmes are reported in Supplementary Table S4.

Signed activity priors. Genes selected by Qwen are converted into signed module weights using their standardized effects in the target dataset. Let $\eta_{cg} = +1$ for an up-regulated marker and $\eta_{cg} = -1$ for a down-regulated marker. The unnormalized and normalized weights are

$$\tilde{w}_{cg} = \eta_{cg} \text{clip}(|d_{cg}|, w_{\min}, w_{\max}), \quad (3)$$

$$w_{cg} = \frac{\tilde{w}_{cg}}{\sum_{h \in \mathcal{M}_c} |\tilde{w}_{ch}|}.$$

The signed, normalized weights encode expected direction and relative within-module contribution, with clipping limiting extreme effects. Let x_{ig}^{std} denote the standardized expression of gene g in profile i . The raw module activity is

$$\tilde{a}_{ic} = \sum_{g \in \mathcal{M}_c} w_{cg} x_{ig}^{\text{std}}. \quad (4)$$

Robust module-wise standardization yields a_{ic} and the activity matrix $\mathbf{A} \in \mathbb{R}^{N \times C}$, providing graded profile-level biological guidance. Each module also receives a reliability coefficient $r_c \in [0, 1]$. Supplementary Equations S5 and S6 detail the bounded multiplicative gates used to estimate r_c from expression separation, exposed-label agreement and biological evidence. Larger r_c values give more reliable modules greater influence during gene prioritization.

2.3. Biology-guided adaptation

Residual adaptation. STCKLA uses the xTrimoGene architecture of scFoundation (Hao et al., 2024). A residual adapter supplies target-specific flexibility to the encoder input. For expression vector $\mathbf{x}_i$,

$$\mathbf{x}_i^{\text{adp}} = \mathbf{x}_i + \alpha \mathcal{A}_\phi[\text{LN}(\mathbf{x}_i)]. \quad (5)$$

Here, LN is layer normalization, $\mathcal{A}_\phi$ is a lightweight nonlinear mapping and α controls the residual update. Figure 2 illustrates how near-identity residual adaptation retains pretrained structure while adapting the encoder input.

Activity-guided gene prioritization. STCKLA uses marker-module priorities to allocate the fixed encoder-visible budget to biologically informative and profile-relevant module genes, reducing the influence of less informative high-expression signals while preserving complementary expression-driven context.

Let $\mathcal{C}_i$ denote the subset of marker modules with the strongest reliability-adjusted activities in profile i . The priority assigned to gene g is

$$q_{ig} = \max_{c \in \mathcal{C}_i: g \in \mathcal{M}_c} |a_{ic}| r_c |w_{cg}|. \quad (6)$$

This product combines profile-level module activity a_{ic} , module reliability r_c and relative gene weight w_{cg} . It ranks module genes that are both biologically supported and active in the current profile, thereby guiding the prior-guided portion of

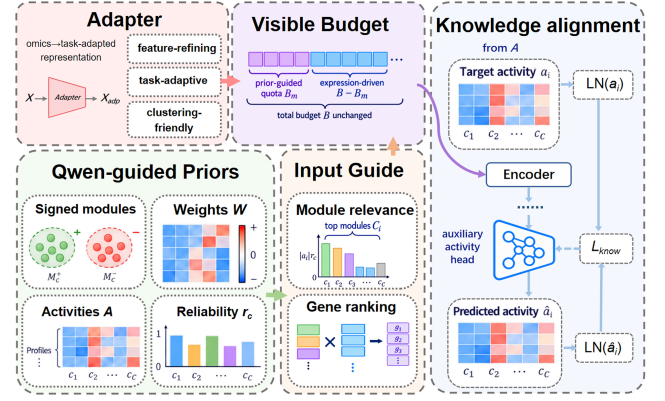


Figure 2 Dual-path integration of Qwen-assisted priors in scFoundation adaptation. The residual adapter refines the omics input before encoder exposure, while Qwen-derived activities and marker weights allocate a fixed visible budget to prior-guided and expression-driven gene tokens. The activity matrix further supplies targets for auxiliary prediction from z_i through Smooth L1 alignment.

the encoder-visible budget. Figure 2 summarizes the two Qwen-prior pathways: fixed-budget gene exposure and continuous activity targets for knowledge alignment.

Clustering embedding. Gene-level encoder output $\mathbf{H}_i$ is pooled into $\mathbf{h}_i$. Supplementary Equation S1 defines the projection and L2 normalization used to obtain $\mathbf{z}_i$. The pooled vector supports label prediction; $\mathbf{z}_i$ is shared by prototype, consistency and knowledge-alignment objectives and is used for clustering.

2.4. Multi-signal representation learning

STCKLA combines reconstruction, supervised classification, prototype learning, consistency regularization and knowledge alignment to preserve expression structure while imposing label, prototype and module constraints.

Reconstruction. The asymmetric decoder predicts designated valid expression positions from encoded context. Supplementary Equation S2 defines the reconstruction loss $\mathcal{L}_{\text{expr}}$ by mean squared error to retain coordinated transcriptomic structure.

Prototype geometry. A classifier attached to the pooled representation produces class probabilities p_{ic} . Supplementary Equation S3 defines the supervised classification loss $\mathcal{L}_{\text{ce}}$ by cross-entropy over exposed labels. STCKLA additionally learns one normalized prototype π_c for each class, assigning labelled profiles by cosine similarity:

$$\mathcal{L}_{\text{proto}} = -\frac{1}{|\mathcal{L}|} \sum_{i \in \mathcal{L}} \log \frac{\exp(\mathbf{z}_i^\top \pi_{y_i} / \tau_p)}{\sum_{c=1}^C \exp(\mathbf{z}_i^\top \pi_c / \tau_p)}. \quad (7)$$

Cross-entropy preserves label-predictive distinctions, whereas prototypes establish shared class anchors and stabilize embedding geometry when labelled examples are sparse or unevenly distributed.

Table 1. Overview of benchmark datasets.

	BRCA	Pan-cancer	4k PBMC	Lung	Liver	Pancreas	Melanoma
Data type	Bulk	Bulk	scRNA-seq	scRNA-seq	scRNA-seq	scRNA-seq	scRNA-seq
Profiles	671	8,314	4,327	10,360	8,444	16,382	4,645
Genes	5,000	3,217	16,655	16,327	20,007	19,093	23,686
Classes	5	32	15	13	20	14	9

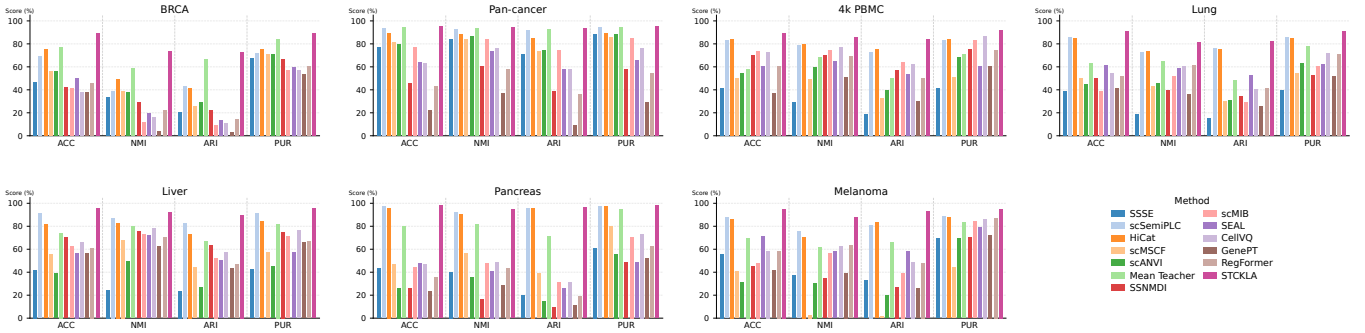


Figure 3 Comparison of clustering performance across seven benchmark datasets. Bars report ACC, NMI, ARI and PUR (%) for all compared methods.

Consistency regularization. Original and identity-preserving perturbed views of each profile are encoded jointly. Supplementary Equation S4 defines the consistency loss $\mathcal{L}_{\text{cons}}$ by cosine consistency between normalized representations, limiting perturbation-driven embedding changes for labelled and unlabelled profiles.

Knowledge alignment. Qwen-assisted modules define the continuous activity vector $\mathbf{a}_i = (a_{i1}, \dots, a_{iC})$, and $\mathcal{H}_\omega(\mathbf{z}_i) = \hat{\mathbf{a}}_i$ predicts it from the clustering representation. Their profile-wise normalized values are aligned by

$$\mathcal{L}_{\text{know}} = \frac{1}{|\mathcal{B}_{\text{know}}|} \sum_{i \in \mathcal{B}_{\text{know}}} \text{SmoothL1}(\text{LN}(\hat{\mathbf{a}}_i), \text{LN}(\mathbf{a}_i)). \quad (8)$$

where $\mathcal{B}_{\text{know}}$ denotes profiles with module-activity targets and includes all profiles in the reported configuration. Profile-wise normalization emphasizes relative activity, and Smooth L1 accommodates graded or overlapping patterns.

Grouped training objective. All reported benchmark experiments used

$$\mathcal{L} = \lambda_{\text{rep}} (\mathcal{L}_{\text{expr}} + \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{proto}}) + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}} + \lambda_{\text{know}} \mathcal{L}_{\text{know}}. \quad (9)$$

The shared coefficient λ_{rep} controls reconstruction, classification and prototype learning, whereas λ_{cons} and λ_{know} control consistency regularization and knowledge alignment. The same default configuration was used across datasets, with implementation-level coefficients and one-factor sensitivity analyses reported in Supplementary Section S5. Reconstruction, consistency and alignment use all profiles, whereas label-guided geometry is learned from $\mathcal{L}$.

2.5. Embedding-based clustering

After adaptation, the residual adapter, selective token procedure, scFoundation encoder and projection head produce $\mathbf{z}_i$ for every profile. Labelled and unlabelled embeddings are clustered jointly by k-means, with K set to the number of classes listed in Table 1.

3. Results

3.1. Evaluation design

Table 1 summarizes the two patient-level transcriptomic datasets and five single-cell RNA-seq datasets used to evaluate STCKLA. These benchmarks cover BRCA molecular subtypes, pan-cancer cohorts, peripheral blood immune cells and tissue-resident cell populations from lung, liver, pancreas and

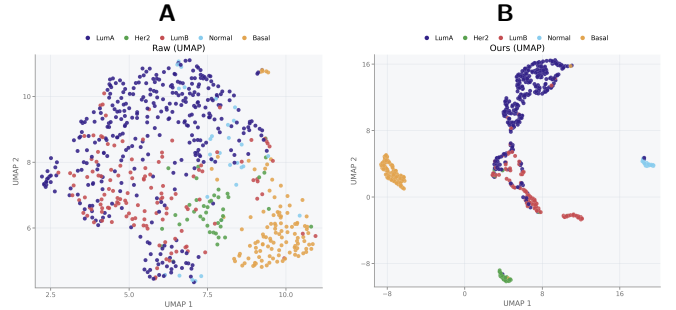


Figure 4 BRCA representations before and after STCKLA adaptation. (A) Raw transcriptomic profiles. (B) Final STCKLA embedding. Points are coloured by reference molecular subtype.

melanoma data. Public data sources and accessions are listed in the Data availability section.

All methods were evaluated on all profiles in each dataset. Clustering performance was assessed using clustering accuracy (ACC), normalized mutual information (NMI), adjusted Rand index (ARI) and purity (PUR). These metrics quantify label agreement, shared partition information, chance-corrected pairwise agreement and within-cluster class homogeneity, respectively. The exact formulae and notation are given in Supplementary Section S1.

3.2. Clustering performance

We compared STCKLA with semi-supervised learning methods, including SSSE (Zeng et al., 2025), scSemiPLC (Ma et al., 2026), HiCat (Bi et al., 2025), scANVI (Xu et al., 2021), Mean Teacher (Tarvainen and Valpola, 2017) and SSNMDI (Qiu et al., 2023); unsupervised clustering methods, including scMSCF (Jiang et al., 2025), scMIB (Yan et al., 2026) and SEAL (Ye et al., 2026); the LLM-derived gene-representation method GenePT (Chen and Zou, 2025); and the biological foundation-model methods CellVQ (Wang et al., 2026) and RegFormer (Hu et al., 2026). The comparator grouping is summarized in Supplementary Table S1.

Figure 3 and Supplementary Tables S2 and S3 show that STCKLA achieved the highest ACC, NMI, ARI and PUR among the compared methods across the seven benchmark datasets, indicating consistent gains in label correspondence, partition recovery, chance-corrected agreement and cluster purity. Figure 4 further shows that, on BRCA, the final STCKLA embedding formed more compact subtype-associated structure than the raw transcriptomic profile space; comparator embeddings are provided in Supplementary Figure S1. STCKLA

Table 2. Qwen-assisted BRCA marker programmes. Complete programmes are supplied in Supplementary Table S4.

Subtype	Positive markers	Negative markers
LumA	<i>ESR1, GATA3, MAPT, DACH1</i>	<i>TUBA1C, UBE2C</i>
Her2	<i>AKT1, MYCN, EPCAM, MMP1</i>	<i>ESR1, IGF1R, KRT14, RARG</i>
LumB	<i>ESR1, GATA3, PCLAF, BUB3</i>	<i>EGFR, KRT14, DEFB1</i>
Normal	<i>EGFR, TXNIP, KIT, JAG1</i>	<i>MCM2, CENPF, FOXM1</i>
Basal	<i>CDC20, AURKB, FOXM1, TOP2A</i>	<i>GATA3, NAT1, ELOVL5</i>

Table 3. Ablation of context-calibrated Qwen priors on BRCA.

Variant	ACC	NMI	ARI	PUR
Static activity control	85.39	66.94	65.55	85.39
Global modules	84.05	70.33	63.09	84.05
STCKLA	89.42	73.30	72.33	89.42

remained effective across the pan-cancer cohort and the five single-cell benchmarks.

3.3. Module contribution

We examined whether Qwen-organized marker modules become usable priors during scFoundation adaptation. Candidate-gene evidence and signed module weights are reported in Supplementary Section S3, and Table 2 lists selected positive and negative markers from the five BRCA programmes. Complete programmes are provided in Supplementary Table S4. We then assessed module use at the input, encoder and activity levels.

Each encoder-visible gene was assigned a gradient-times-input contribution score for the prototype decision margin (Simonyan et al., 2014; Ancona et al., 2018). Figure 5A shows stronger contribution enrichment for Qwen-selected module genes than for Qwen-rejected and data-ranked candidates. Figure 5B further shows increasing alignment between module activity and deeper scFoundation encoder layers by representational-similarity analysis (Kriegeskorte et al., 2008), whereas permuted activities remained near zero. Figure 6 shows that class-specific module activities were also elevated in their matched BRCA subtypes. These analyses indicate that Qwen-organized modules are reflected in feature use, encoder geometry and subtype-resolved activity.

Table 3 shows that both controls reduced performance across all metrics. The global-module control tests class-resolved Qwen organization, whereas the static-activity control tests profile-resolved calibration. Their declines indicate that Qwen-derived marker knowledge becomes useful through both class-specific module construction and sample-specific activity calibration; the component ablations below examine how these calibrated priors are coupled to representation learning.

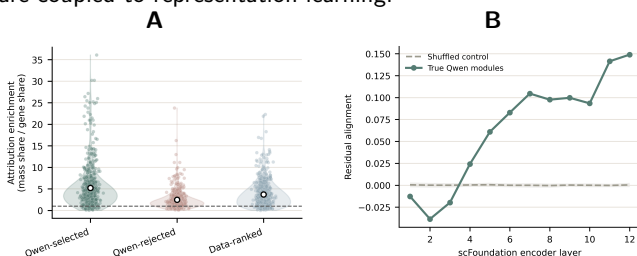


Figure 5 Downstream use of Qwen-assisted BRCA marker modules. (A) Contribution enrichment. (B) Layer-wise module-encoder alignment.

Table 4. Ablation analysis on BRCA.

Variant	ACC	NMI	ARI	PUR
STCKLA	89.42	73.30	72.33	89.42
w/o module-guided input selection	88.82	70.77	71.93	88.82
w/o knowledge alignment	81.67	66.44	58.86	81.67
w/o residual feature adapter	78.54	51.48	53.33	78.54
w/o expression reconstruction	87.93	70.65	70.54	87.93
w/o supervised classification	88.08	71.04	70.11	88.08
w/o prototype learning	59.76	51.81	34.96	79.43
w/o cross-view consistency	79.58	57.91	53.22	81.22

3.4. Ablation analysis

We then used BRCA component ablations to examine how the calibrated Qwen prior informs representation learning, alongside the shared representation-learning objectives. Module-guided input selection, knowledge alignment, the residual feature adapter, expression reconstruction, supervised classification, prototype learning and cross-view consistency were each removed while retaining the remaining components.

Table 4 reports the best ACC, NMI, ARI and PUR for STCKLA, with the corresponding bar-plot visualization provided in Supplementary Figure S3. The reductions after removing module-guided input selection or knowledge alignment show that input prioritization and activity-based supervision are complementary routes through which calibrated Qwen priors inform encoder exposure and embedding geometry. Together with Table 3, these ablations provide a staged account: class-resolved organization and profile-resolved calibration make Qwen modules usable, while input guidance and knowledge alignment couple them to representation learning.

The residual feature adapter, expression reconstruction, supervised classification and consistency regularization also contributed to clustering quality. The largest decline followed removal of prototype learning, reducing ACC from 89.42 to 59.76 and ARI from 72.33 to 34.96. Thus, Qwen-organized priors act alongside the shared objectives that shape clustering geometry.

Detailed one-factor hyperparameter sweeps are provided in Supplementary Section S5.

3.5. Label-rate analysis

To assess the dependence of STCKLA on labelled profiles, we varied the BRCA label proportion from 1% to 40% while keeping the remaining configuration fixed. All profiles were included during training and evaluation, and ACC, NMI, ARI and PUR were computed for the final partitions. Figure 7 shows that performance increased sharply from 1% to 20% labelled profiles, with ACC rising from 33.23 to 89.42 and ARI from 6.63 to 72.33. Increasing the labelled fraction from 20% to 40% did not yield a uniform additional gain, indicating a plateau after sufficient label information had been provided. The corresponding numerical values are provided in Supplementary Table S6.

4. Discussion

STCKLA adapts scFoundation for semi-supervised transcriptomic clustering by converting Qwen-assisted biological reasoning into signed marker modules, profile-specific gene priorities and continuous module-activity targets. Across two patient-level and five single-cell benchmarks, it achieved the best overall ACC, NMI, ARI and PUR among the evaluated methods. Controlled analyses showed that class-resolved module

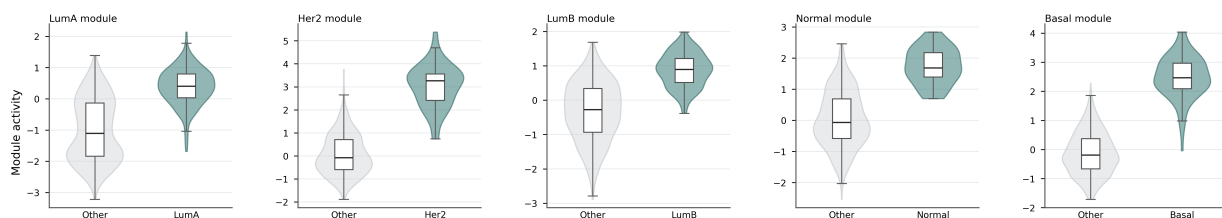


Figure 6 Subtype-specific activity of Qwen-assisted BRCA marker modules.

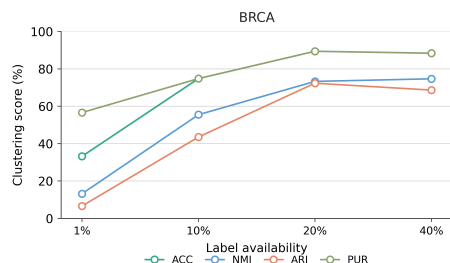


Figure 7 Effect of label availability on BRCA clustering performance. ACC, NMI, ARI and PUR are shown for STCKLA using 1%, 10%, 20% and 40% labelled profiles.

organization, profile-specific activity, prototype learning and knowledge alignment each contributed to performance; in BRCA, prototype removal produced the largest degradation.

These findings position context-specific biological priors as quantitative constraints for representation adaptation. They also indicate a general design principle for LLM-assisted omics analysis: language-model reasoning becomes a useful biological prior when anchored in cohort-specific candidate signals, organized with biological and class context, calibrated at the profile level, and coupled to representation learning. The current evaluation is limited to the selected datasets, evidence sources and Qwen backbone, and future work should examine module transferability, alternative LLM backbones and independent molecular validation. By coupling scFoundation with Qwen-assisted biological priors, STCKLA provides a practical route to task-adapted and biologically interpretable transcriptomic clusters.

5. Conflicts of interest

The authors declare no competing interests.

6. Funding

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7. Data availability

The BRCA and pan-cancer datasets analysed in this study were obtained from MLOmics (Yang et al., 2025; Kotoge, 2025) through the MLOmics Figshare record. The 4k PBMC data are available from the 10x Genomics 4k PBMC reference dataset (10x Genomics, 2017). The remaining single-cell benchmarks were obtained from the original studies and from the public GEO or ArrayExpress accessions listed here: lung from the human lung cell atlas study (Vieira Braga et al., 2019) (GSE130148), liver from the human liver single-cell study (MacParland et al.,

2018) (GSE115469), pancreas from the integrated pancreas single-cell atlas (Luecken et al., 2022): GSE81076, GSE85241, GSE86469, GSE84133, GSE81608 and E-MTAB-5061, and melanoma from the melanoma single-cell study (Tirosh et al., 2016) (GSE72056). Source code, model configurations, Qwen prompt templates and files required to reproduce the reported analyses are available at <https://github.com/III1206/STCKLA>.

8. Author contributions statement

Jiale Liu, Zheng Chen and Jinli Zhang conceived the study. Jiale Liu developed the methodology and software. Jiale Liu performed the experiments and analysed the data. Ziwei Yang and Jiale Liu curated the datasets. Jiale Liu wrote the original draft. Jinli Zhang, Zheng Chen and Junzhe Jiang reviewed and edited the manuscript. Zheng Chen, Jinli Zhang and Zongli Jiang supervised the study. All authors approved the final manuscript.

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